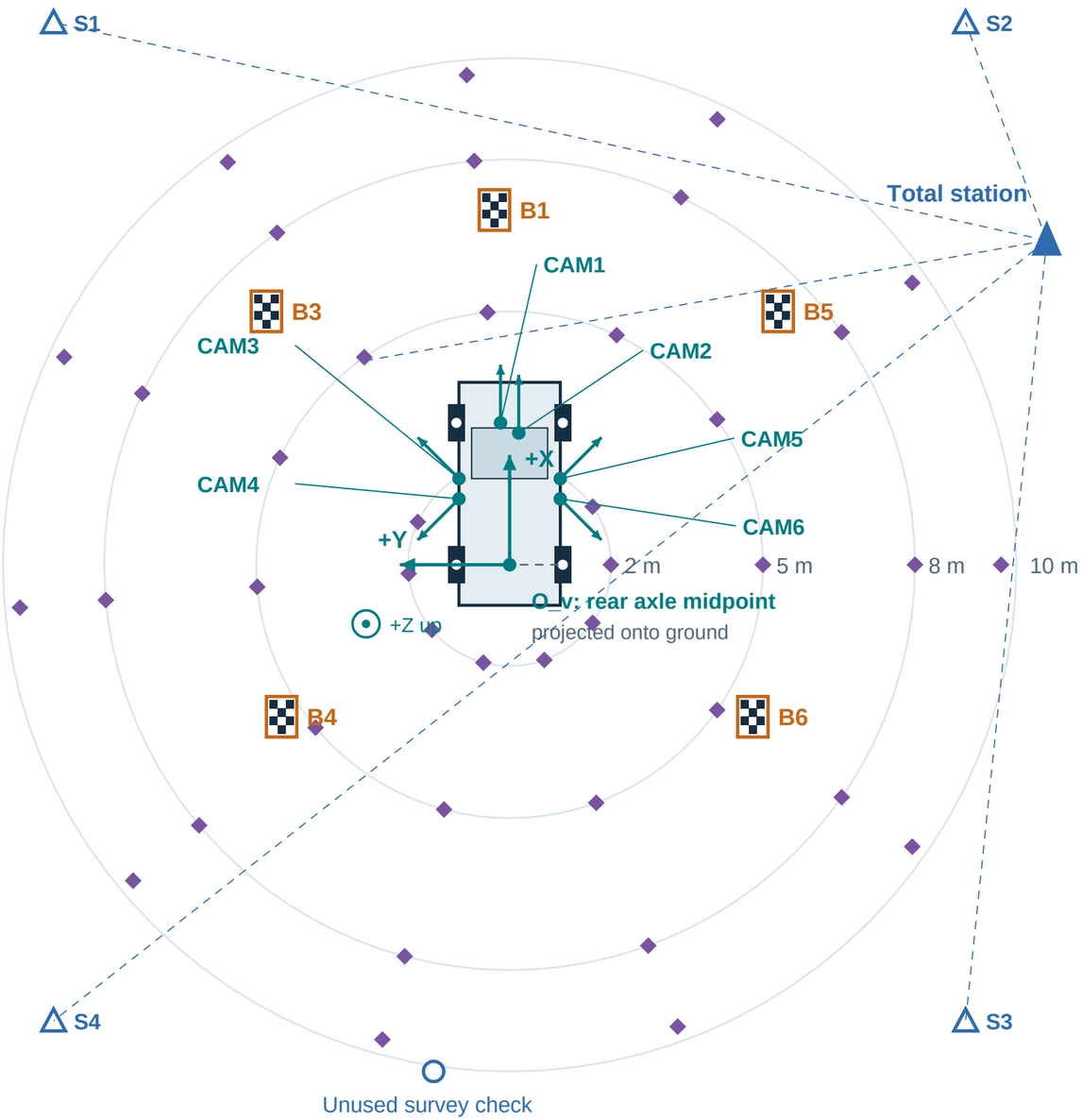
One vehicle frame. Six independently surveyed cameras.

Overhead planning layout | Range bands are centered on the vehicle origin, not on each camera.



Board icons = example locations; each camera needs at least six varied placements.

Purple points = candidate ground checks. Visibility must be confirmed in real images.

CURRENT CAMERA INVENTORY

ID	Schematic direction	FOV label
CAM1	Forward / narrow	40 deg
CAM2	Forward / wide	110 deg
CAM3	Front-left	70 deg
CAM4	Rear-left	70 deg
CAM5	Front-right	70 deg
CAM6	Rear-right	70 deg

FOV labels have no confirmed axis type.
Arrows show direction only. Camera positions,
yaw, pitch, roll and vehicle size are illustrative.

THREE DISTINCT MEASUREMENT LAYERS

Survey network + wheel references

At least four stable controls; redundant checks.
Survey all four hub centers and the ground.

Checkerboards for extrinsic fitting

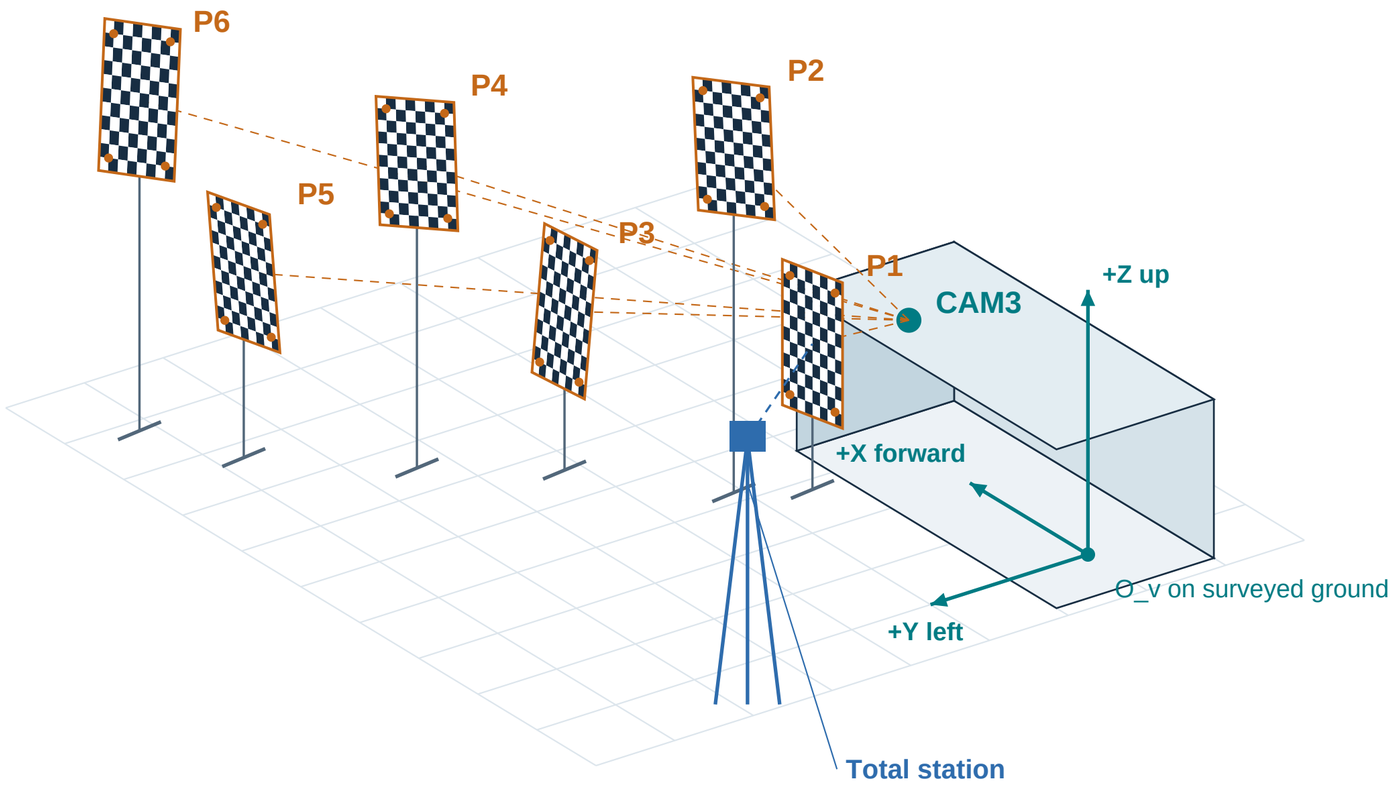
Survey O/A/B/C at every stationary placement.
Tie each camera to the same vehicle frame.

Independent ground validation

Aim for 24 observations per camera.
About six per observable range band;
test near 10 m where actually visible.

Move the surveyed board through a 3D volume.

Example for CAM3 | Six sequential placements of one movable board, not six required physical boards.



Schematic perspective: dimensions, camera height, poses and target positions are illustrative.

All four extreme inner corners (or verified offset fiducials) are surveyed at each placement.

PLACEMENT DESIGN

At least six placements per camera.
Use two or more distances, several directions, varied heights, depths and board tilts.
Keep all 7 x 10 inner corners readable.

FOR EVERY PLACEMENT

1. Identify board + placement.
2. Fix the board; survey O/A/B/C.
3. Correct prism and fixture offsets.
4. Capture several sharp, static images.
5. Recheck a control for board motion.
6. Approve O/A/B/C corner ordering.

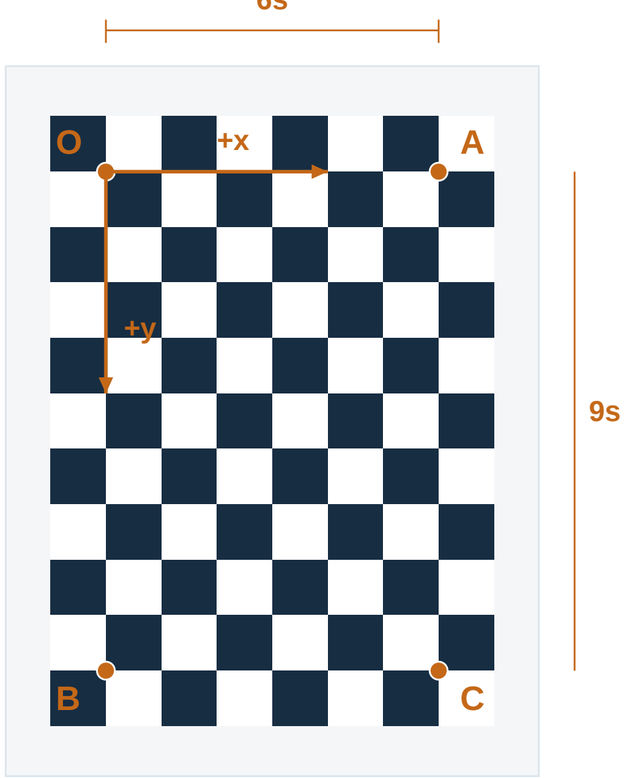
ONE FIXED CAMERA POSE

Combine the distributed placements to estimate one camera-to-vehicle transform.
Camera overlap is optional; a shared survey frame provides the connection.

Measure the right corners. Preserve the frame conventions.

Checkerboard reference, wheel-derived vehicle frame, and the six extrinsic parameters.

A / SURVEYED CHECKERBOARD



8 x 11 squares -> 7 x 10 inner corners

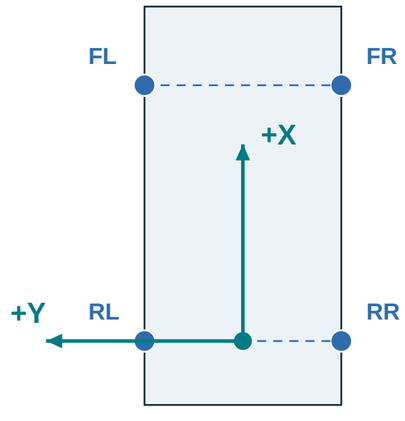
O=(0,0,0) A=(6s,0,0)

B=(0,9s,0) C=(6s,9s,0)

s = measured square spacing in meters

Viewed from the printed face: +z points into the board. Survey inner corners, not paper edges. Printed size and measured spacing remain inputs.

B / VEHICLE FRAME



O_v at ground level

1. Survey four actual hub-center references.
2. Survey and check the ground plane.
3. +Z = upward plane normal.
4. +X = rear-to-front axle-midpoint vector, projected onto the ground plane.
5. +Y = +Z cross +X; orthonormalize.

Origin: rear hub midpoint projected onto the surveyed ground plane.
Correct locating-fixture offsets first.

The inset is a top view; hub heights are omitted.

C / CAMERA EXTRINSICS

Translation: tX, tY, tZ [meters]

Rotation: 3 orientation degrees of freedom (yaw, pitch, roll; define order if exported)

T_vehicle_from_camera

p_vehicle = R p_camera + t

The inverse maps vehicle to camera.
Optical axes: x right, y down, z forward.
Use the same units and explicit frame names.

INDEPENDENT ACCEPTANCE

e_XY <= 0.05 m

Use separate surveyed ground points.
Range r = sqrt(X^2 + Y^2) from O_v.
Bands: 0-2, >2-5, >5-8, >8-10 m.
Only claim the usable, sampled ground region.

No claimed pass from these drawings.
Missing far coverage -> NOT VALIDATED.
A failed point remains a failure.
Report survey uncertainty with results.